Complete Task List: Electricity Anomaly Project

Status as of Monday 2 November 2026 (evening). Final deadline: 17 November 2026. Task IDs follow the Project Task Tracker and Folder Plan. Status: **Done**, **Partly done**, **TO DO**.

47 tasks: 31 done, 1 partly done, 15 to do. Optional items, dropped first if time runs short: T23, T41, T42, T43.

Phase 1. Proposal, data and first look

ID	Task	Status	Where
T01	Write and submit the proposal	Done	Proposal documents
T02	Get the data (GEPIII, LEAD)	Done	ashrae-energy-prediction/
T03	First look at the data	Done	archive/old_notebooks
T04	Partner's all-buildings cleaning and prediction run	Done	Mid-progress.ipynb (archive; do not quote)

Phase 2. The midterm pipeline

ID	Task	Status	Where
T05	LEAD labels, 200 buildings, 5,811 events	Done	notebook 01
T06	Check LEAD against GEPIII, site 0 to kWh, zero runs	Done	notebook 01
T07	Data-quality checks on GEPIII	Done	notebook 02
T08	Set the time split	Done	config.py
T09	Hourly grid, features, lags	Done	notebook 03
T10	Two baselines (last hour, typical value)	Done	notebook 03
T11	Train models A, B, C	Done	notebook 03
T12	Per-building scores and one cut-off per method	Done	notebook 04
T13	Flag the test period, fixed percentile	Done	notebook 04

ID	Task	Status	Where
T14	Evaluate flags against events	Done	notebook 05
T15	Compare methods at equal flag rates	Done	notebook 06
T16	Midterm report, slides, script	Done	Docs, slides
T17	Midterm presentation (20 Oct 2026)	Done	n/a

Phase 3. Setting up the final project

ID	Task	Status	Where
T18	Split the combined notebook into notebooks 01 to 07	Done (run and verified 2 Nov)	notebooks/
T19	Explanation documents (plain English, master, plan, reference)	Done	Docs
T20	Run notebooks on the real data and verify	Done: data parts exact, models close, ranking unchanged, clean rerun identical	notebook 07
T21	Correct older documents	Done	Docs, LaTeX, markdown
T22	Settle open decisions	Done (test protection option A assumed, please confirm)	docs/decision.md

Phase 4. Improving the detector and the evidence

ID	Task	Status	Where
T23	More history clues (lags 2, 3, 24-hour average)	TO DO (optional)	planned
T24	Persistence test: does history make the model miss long anomalies?	**Done: copying the last hour loses long events (32.7% vs Model C 50.0%); Model C does not**	notebook 09, persistence_test.png

ID	Task	Status	Where
T25	Isolation Forest	Done: worse than last hour and Model C	notebook 08
T26	Label-guided per-building cut-off	Done: tested, kept the single pooled cut-off	notebook 08, eval_frozen_rules_test.csv
T27	Delay and hours inside events at equal rates	Done	notebook 08
T28	Bootstrap intervals grouped by building	Done: Model C and last hour tied	notebook 08
T29	Read the unmatched flags by hand (DONE 2 Nov: pack of 30 built and rated by Claude from the plots, all 714 unmatched runs tagged, notebook 10; the team should cross-check the ratings)	Done	planned
T30	Ranked review list	TO DO	planned
T40	Retrain without labeled hours and compare	Done: Model C 38.0% vs 35.4%	notebooks/label_excluded
T38	Site-wide explain-away filter	TO DO	planned
T39	Permutation importance for A, B, C (interpretability)	TO DO	planned

Phase 5. Third dataset

ID	Task	Status	Where
T31	Prepare CU-BEMS (download, hourly, split)	Done (9 Nov)	Notebook 11; data/cu_bems_hourly/. Open check: 2019Floor6.csv ends 22 Oct 2019 (may be an incomplete download)

ID	Task	Status	Where
T32	Plant anomalies and evaluate	Done (9 Nov)	Notebook 12; results/cu_bems/. Planted in test only, 20 placements; Model C′ finds 41% at 2% flagged vs 32% last hour

Phase 6. Extras (drop first)

ID	Task	Status
T41	Small tuning pass for Model C on validation	TO DO (optional)
T42	Sensitivity of zero-run length, MAD floor and cut-off target	TO DO (optional)
T43	Results by building type and by site	TO DO (optional)

Phase 7. Writing and handing in

ID	Task	Status
T33	Final tables and figures	TO DO
T44	Map the report to the AIT thesis template	TO DO
T45	Related works and citations	TO DO
T46	Discussions: performance, interpretability, complexity	TO DO
T47	Honest comparison with other approaches (same-as-last-hour ties Model C)	TO DO
T34	Final report	TO DO
T35	Final slides and speaking script	TO DO
T36	Tidy and package the folder (requirements.txt is written; still to do: delete _rerun_tmp, final packaging)	Partly done
T37	Final presentation	TO DO

Open item

Confirm with Arpan that the test period is scored once with frozen settings (option A).